

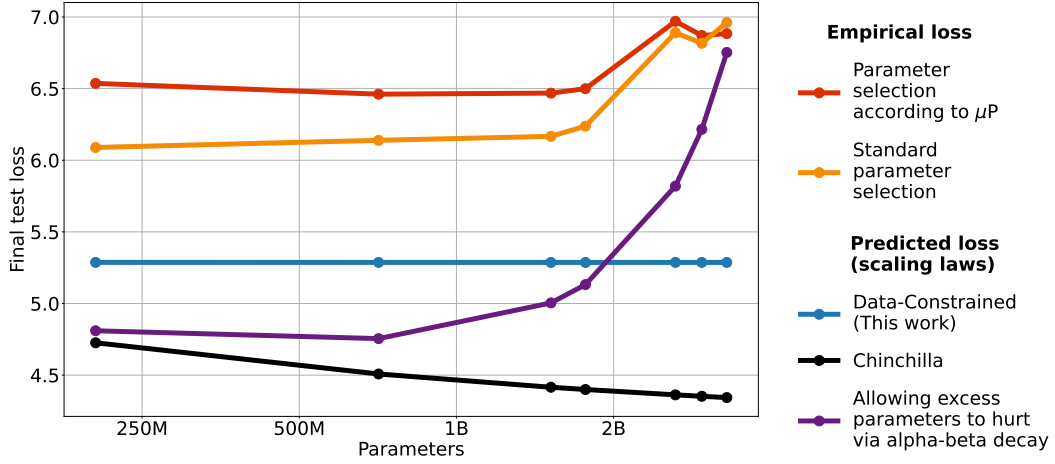


Figure 11: **Empirical and predicted losses of LLMs trained on 100 million tokens for a single epoch.** Excess parameters empirically hurt performance, but this may be due to a lack of regularization. Thus, our scaling formula predicts loss to plateau, while Chinchilla predicts loss to improve. By decaying the exponent α (and β) instead, one can allow excess parameters to hurt.

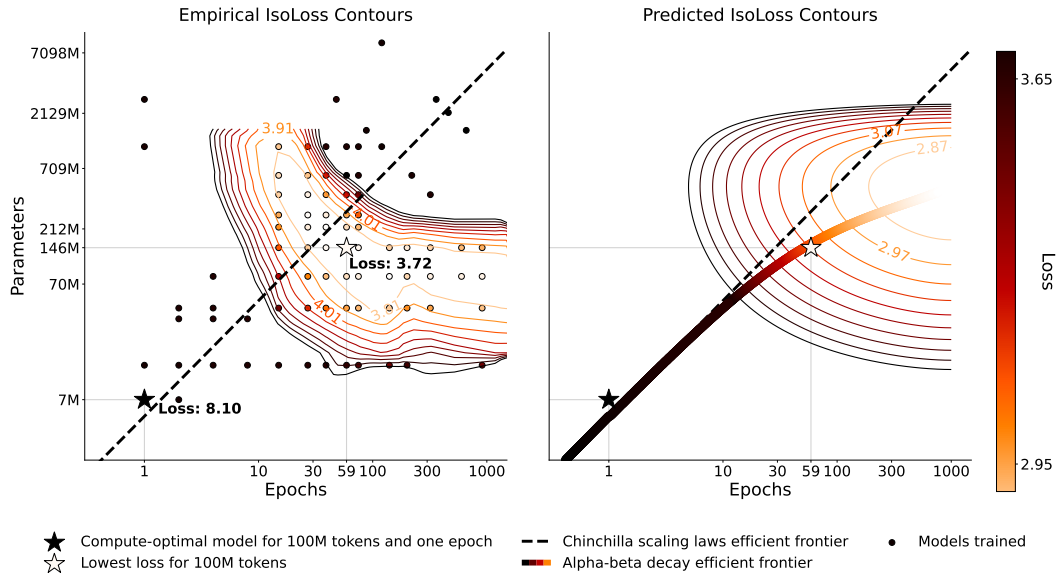


Figure 12: **IsoLoss contours for 100 million unique tokens with contours predicted by parametric decay of alpha and beta.** The same models from Figure 3 with the contour predictions being done by the alpha-beta decay formulation introduced in Appendix F.

G Case Study: Galactica

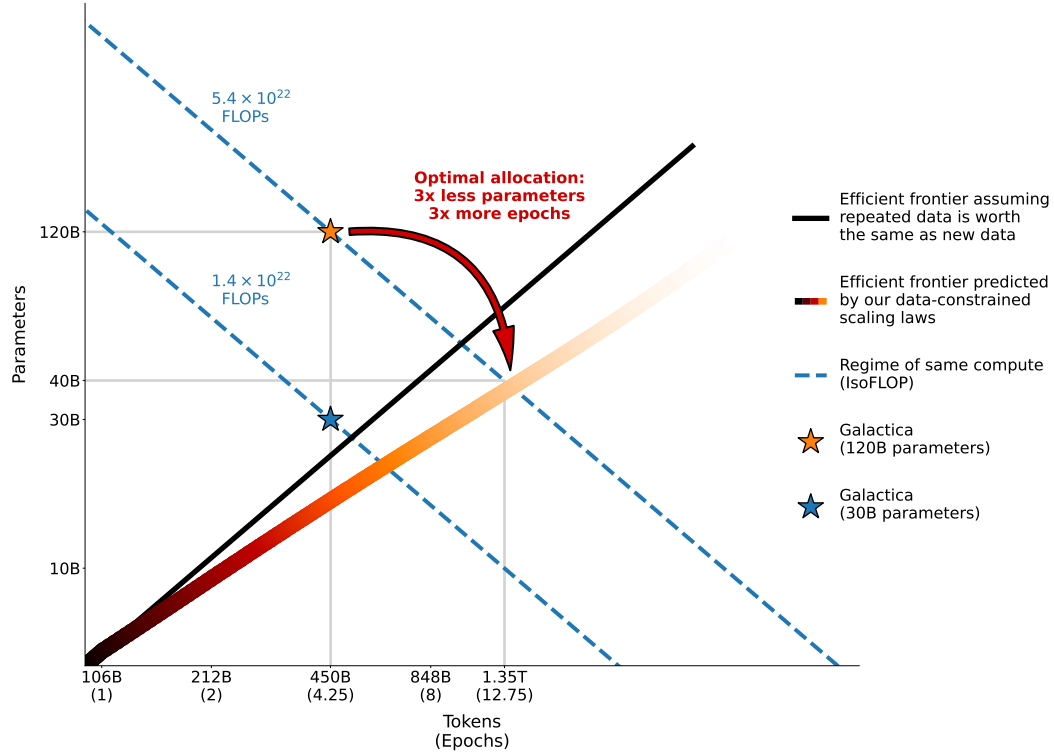


Figure 13: **Optimal compute allocation for Galactica.** Efficient frontier assuming repeated data is worth the same as new data (Chinchilla scaling laws) and data-constrained efficient frontier assuming a unique token budget of 106 billion tokens like for the Galactica models [108]. For optimal compute allocation according to our proposed data-constrained scaling laws, the 120 billion Galactica model should have been significantly smaller and trained for more epochs.

The Galactica models [108] are the only publicly known LLMs that explicitly trained for a significant number of epochs prior to this work. They trained their models on 106 billion unique tokens for 4.25 epochs. Our findings on *Return* from repeated data agree with their conclusion that multiple epochs are beneficial, however, we find that even more epochs can be beneficial and a small spike in validation loss does not justify stopping training (Appendix J). Meanwhile, our findings on *Allocation* significantly deviate from Galactica. Figure 13 visualizes the Galactica models with our predicted efficient frontier in the same style as Figure 1. The creators of Galactica decided to train a 120 billion parameter model on 450 billion tokens, a significant overallocation to parameters even in Chinchilla terms (black efficient frontier). This decision was likely driven by the intuition that repeated data is worth less, thus one should spend more compute on parameters. However, our empirical data contradicts this. Parameters learning from repeated data are worth *even less* than repeated data, thus one should overallocate to epochs, not parameters. Our data-constrained scaling laws thus predict that a better model could have been trained by allocating significantly more FLOPs to epochs rather than parameters for the largest Galactica model with 120 billion parameters. Specifically, 40 billion parameters trained for 1.35 trillion tokens (12.75 epochs) would have been optimal according to data-constrained scaling laws. Note that these scaling laws have been fitted on C4, which is not the dataset used to pre-train Galactica. The Galactica models are pre-trained on a predominantly scientific dataset, which includes code data among other data sources. Results from [42] show that there are differences in the scaling coefficients when training on C4 as compared to GitHub code, however, the overall allocation trend is the same. Thus, while we expect a smaller model trained for more epochs to be better than the 120 billion parameter model, the optimal allocation is unlikely to be exactly 40 billion parameters and 1.35 trillion tokens.

H Training Loss

Hoffmann et al. [42] use training loss as their core metric. However, when repeating data for multiple epochs, training loss is a bad metric as models will overfit to the limited data available as shown in Figure 14. Thus, we use loss on a held-out test set as our key performance metric.

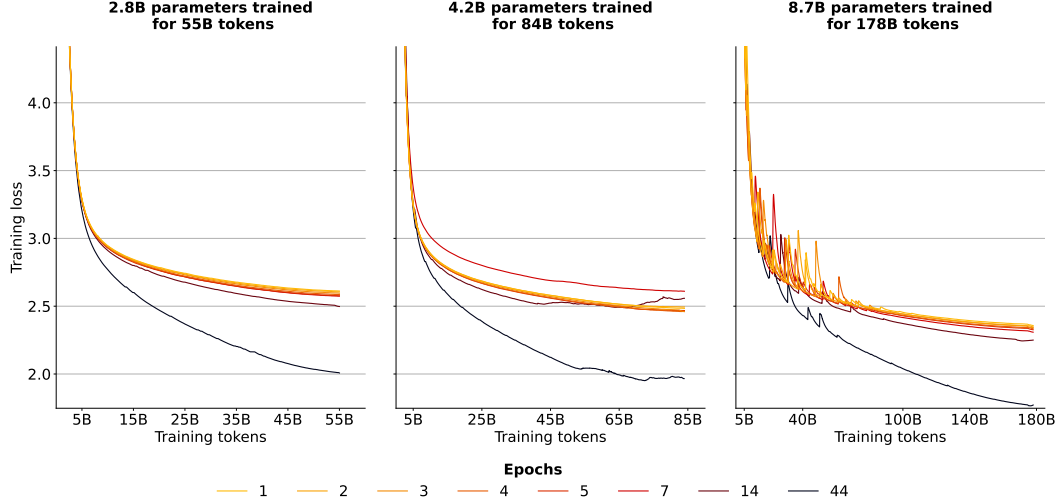


Figure 14: **Training loss smoothed with exponential moving average smoothing and a weight of 0.999.** Models trained on fewer unique tokens (more epochs) have better training loss as they overfit.

I Scaling Curves on the OSCAR Corpus

To ensure our findings are not dataset-dependent, we train models with the same configurations from Figure 4 on the OSCAR corpus [83]. OSCAR is considered noisier than C4 [90] due to its less stringent duplication. Figures 15, 16 depict the validation and training loss of these models. We find the trend to be the same as for models trained on C4: While models with fewer repeats have better loss, differences for a few repeats are insignificant.

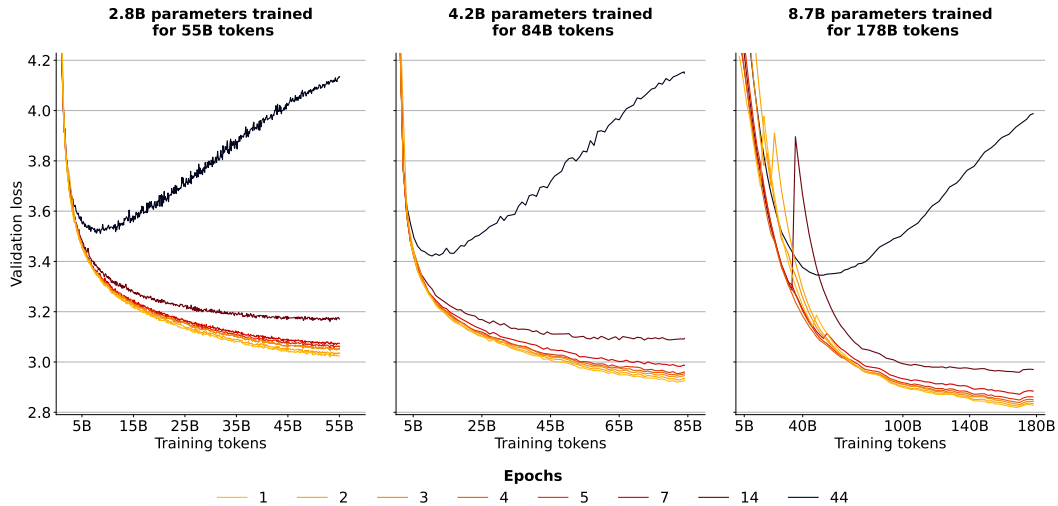


Figure 15: **Validation loss during training for models trained on OSCAR.** Models trained on tokens that are repeated for multiple epochs have consistently worse loss.